ASE 420 Tools Installation - VSCode

VSCode is Your Weapon of Choice

In the age of AI, many IDE tools have emerged, but VS Code is the first tool that software engineers should master.

Install VS Code

- Download VS Code | code.visualstudio.com
- After installing, open VS Code once to confirm it launches correctly before continuing.

VS Code Extensions

- We use Visual Studio Code as the primary IDE.
    - Install the Python extension.
    - Install any other extension needed for your project.

Why VS Code for ASE 420?

- **Python support:** first-class interpreter selection, linting, and IntelliSense.
- **Jupyter:** run notebooks without leaving the editor.
- **Debugger:** set breakpoints and step through code visually.
- **Git integration:** clone, commit, and review changes in the same window.

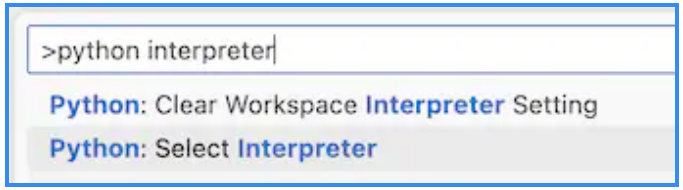
Run the Tetris programs

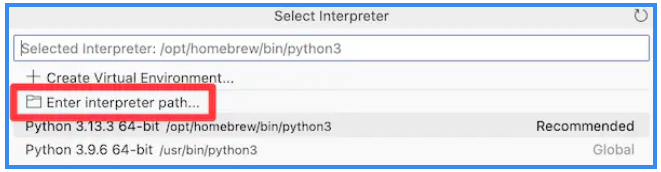
- Clone the ASE420 course repository and open the Tetris example.

```
git clone https://github.com/nkuase/ase420
cd ase420/course_projects/code
pip install pygame
python tetris_ver1.py
```

Step 1: Select the Python Interpreter

- Install the Python extension in VS Code.
- Open the Tetris program in VS Code.
    - Select the interpreter from the project's virtual environment (status bar, or the `Python: Select Interpreter` command). See VS Code docs.





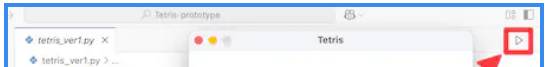
Step 2: Run the App

- Click the arrow to run the app, or run it from the terminal.
- **Recommended:** install `pygame` with UV.

```
uv add pygame
uv run tetris_ver1.py
```

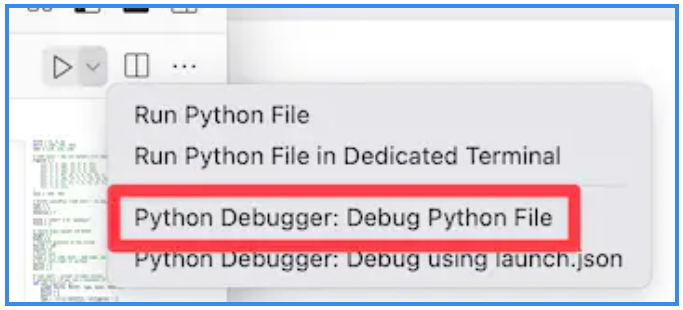
- **Alternative:** install `pygame` with `pip` in your `venv`.

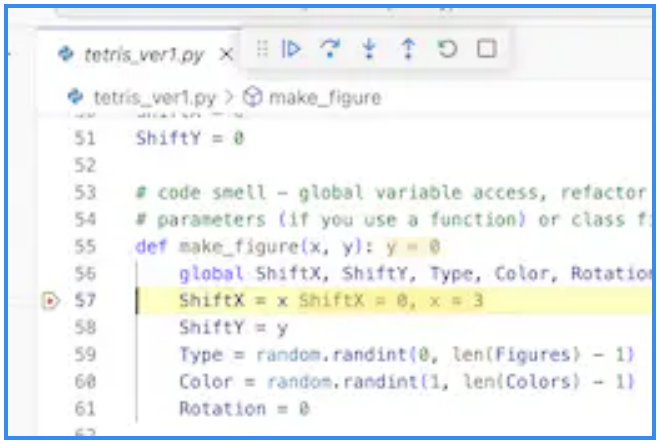
```
pip install pygame
python tetris_ver1.py
```



Step 3: Debug with Breakpoints

- Use VS Code: choose the Python debugger and set breakpoints.





Use Jupyter

What Problems Does Jupyter Solve?

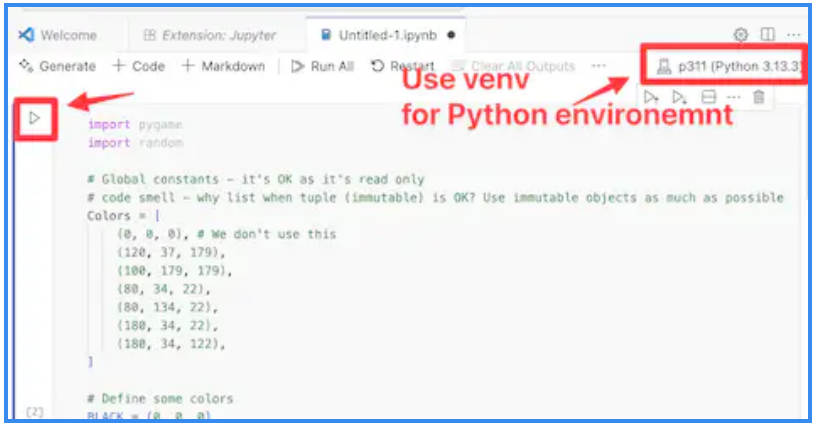
1. Combine code, explanation, and results
2. Develop and test code interactively

What Benefits Does Jupyter Give?

- Combines code, output, and explanation in one notebook, so notes and results stay next to the code that produced them.
- Runs and re-runs small pieces of code (cells) independently, without restarting the whole program.
- Well suited for exploring and visualizing intermediate results, such as testing Tetris logic step by step.

- Install Jupyter extension & set Python environment as your venv Python.

- Use Jupyter to learn Python and analyze the Tetris program.



Use Profile

What Problems Does Profile Solve?

- Switching projects manually reconfigures extensions and settings, which is slow and error-prone.
- A **Profile** saves and switches named collections of extensions, settings, and layouts for each context (e.g., Python vs. web).

What Benefits Does Profile Give?

- **Isolation:** Course-specific settings don't leak into your personal or other project setups.
- **Fast switching:** Move between contexts (e.g., "ASE420", "Web Dev") with one click instead of reconfiguring VS Code.
- **Shareable:** Export a profile so teammates or students start with the same setup.

RECOMMENDED

Create an ASE420 Profile

- Create a dedicated VS Code profile for this course containing the Python extension, debugger, and Jupyter extension.
- This keeps your course environment isolated and easy to reset if something goes wrong.

Use Workspace

What Problems Does Workspace Solve?

- Opening project folders separately breaks search, IntelliSense, and settings continuity.
- A **multi-root workspace** combines multiple related folders into a single VS Code session.
- Use it when your project spans multiple folders; a single-folder project usually doesn't need one.

What Benefits Does Workspace Give?

- **Unified search:** Search and navigate across all folders at once.
- **Consistent settings:** Share extensions, settings, and debug configurations across the whole workspace.
- **Fewer window switches:** Work on related repositories side by side in one window.

Example: Multi-root Workspace

```
{
  "folders": [
    {
      "path": "./course_projects/src"
    },
    {
      "path": "./course_projects/code"
    }
  ]
}
```

Use Copilot Student

NKU students verified through the **GitHub Student Developer Pack** get free access to **GitHub Copilot Student**.

What Problems Does Copilot Solve?

- Writing boilerplate and repetitive code by hand slows development.
- Looking up syntax or API details in documentation interrupts your flow.
- **GitHub Copilot** solves this with AI-based code completion built directly into the editor.

What Benefits Does Copilot Give?

- **Faster typing:** Auto-completes repetitive code such as loops, boilerplate, and common patterns.
- **Inline help:** Suggests function usage and syntax without leaving the editor.
- **Free for NKU students:** Verify through the GitHub Student Developer Pack to get GitHub Copilot Student at no cost. (Plan details and availability can change — check the current GitHub Education page.)

CAREFUL

Focus on the problem, not the tool

- Copilot suggestions are not always correct; always read and understand code before accepting it.
- You are still responsible for the code you submit, including anything Copilot generated.
- Use Copilot to move faster, not to skip understanding the problem.